

Problem 4: Dynamic Weighted Voting System (DWVS) From Bobby Bones Controversy to Fair System

Design Objectives (4 Pillars)

Entertainment

High fan
engagement
in early weeks

Expertise

Technical
skill matters
in finals

Fairness

Prevent
low-skill wins
via fan mob

Transparency

Clear rules
explainable
to audience

Core Formula

$$S_i(w) = \alpha(w) \cdot J_i + [1 - \alpha(w)] \cdot F_i \cdot T_i$$

where: $S_i(w)$ = Comprehensive score | J_i = Judge score % | F_i = Fan vote %
 $\alpha(w)$ = Dynamic judge weight | T_i = Threshold penalty factor

Innovation 1: Dynamic Weight Mechanism

$$\alpha(w) = \min(0.30 + 0.04w, 0.80)$$

Evolution Table:

Week | $\alpha(w)$ | Judge% | Fan% | Philosophy

1	0.34	34%	66%	Entertainment
2	0.38	38%	62%	driven
5	0.50	50%	50%	(Audience
9	0.66	66%	34%	building)
10+	0.74	74%	26%	Merit-based

Early: High fan weight (engagement)

Finals: High judge weight (technical)

Parameter Optimization: Grid Search

Search Space:

- Base α $\in$ [0.25, 0.40] (step=0.01)
- Increment $\Delta\alpha$ $\in$ [0.02, 0.06] (step=0.01)
- Total: 80 combinations

Optimal Result:

- $\alpha = 0.30$ (base judge weight)
- $\Delta\alpha = 0.04$ (weekly increment)
- Score = 0.98-0.99 (near-perfect)

Robustness:

Stable across ± 0.05 parameter range
Diagonal pattern shows consistency

Innovation 2: Threshold Penalty

- 0.3 if $J_i < 0.5 \times J_{\text{week}}$

$T_i =$

- 1.0 otherwise

Purpose:

- Prevent "low-skill high-popularity"
- If judge score < 50% average:
→ Fan votes count only 30%
- Maintains baseline quality standard

Bobby Bones Example (Week 9):

- Judge score = 45% < 50% threshold
- Fan votes: 35% → 10.5% ($\times 0.3$)
- Cannot accumulate enough to win

Result: Proportional correction,
not arbitrary punishment

Impact Assessment: 32 Controversial

Overall Statistics:

- 78.1% rank LOWER under DWVS (25/32)
- Average adjustment: +0.45 positions
- Correlation (Score vs Change): $r=0.42$
→ Proportional, not arbitrary!

Top Cases:

Contestant | Season | Original | DWVS | Change

Bobby Bones	27	1st	2nd	+1
Bristol Palin	11	3rd	4th	+1
Jerry Rice	2	2nd	3rd	+1
Patricia Richardson	5	5th	6th	+1

System Comparison: Pareto Improvement

System | Fairness | Expertise | Balance | Engagement | Overall | Rank

DWVS	5/5	4/5	5/5	4/5	4.5	1st
Current	3/5	3/5	3/5	4/5	3.25	2nd
Judge-only	3/5	5/5	3/5	2/5	3.25	3rd
Fan-only	2/5	1/5	2/5	5/5	2.50	4th

DWVS = Only system achieving 5/5 on both Fairness and Balance

Key Insight: Bobby would place 2nd,
not 1st → Controversy prevented